SIMATS ENGINEERING

ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1506

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 25**
Question Count: 5

Code of Conduct

I M Phanith Chandu (192425238) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

M Phanith Chandu

Assessment Tasks

Short Answer Test

SET 1

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

CO-1 - ASSESSMENT TOOL - 3

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CS41506 - Cloud computing and Big Data Analytics

(1) Cloud Computing is a model for enabling convenient, on-demand network access to a shared pool of configurable computing resources that can be rapidly provisioned and released with minimal management effort or service provider interaction.

i) On-demand self service : Users can access computing resources whenever required without contacting the service provider.

ii) Broad network access : Cloud services are accessible over the internet through devices.

iii) Resource pooling : Computing resources are shared among multiple users using a multi-tenant model.

iv) Rapid elasticity : Resources can increased or decreased quickly according to demand.

(2) mainframe computing → Personal computers → client server computing → Distributed computing → Virtualization → Internet and web applications → Utility computing and cloud computing.

* Faster Networks, powerful hardware, virtualization, distributed storage, automation, and web technologies enabled platforms, such as AWS, Azure, and Google Cloud to provide scalable, reliable, and cost-effective services.

(1) Mainframe Computing :- Large centralised computers served multiple users through terminals.

(2) Personal Computers :- Computing became more accessible to individuals and organisations.

(3) Client - server Computing :- Clients requested services and data from centralised servers over networks.

(4) Distributed Computing :- Multiple computers worked together to perform tasks and share resources.

(5) Virtualisation :- A single physical server could run multiple virtual machines, improving hardware utilization.

(6) Internet and web applications :- Internet connectivity enabled users to access software and services remotely.

(7) Utility Computing and Cloud Computing :- Computing resources became available on demand, with usage - based pricing and scalable infrastructure.

(3) Server virtualization is a technology that divides a physical server into multiple independent virtual machines using software called a hypervisor. Each VM can run its own operating system and applications.

Virtualization	Cloud Computing
i) A technology that creates virtual versions of computing resources	i) A service model that delivers computing resources over a network
ii) Focuses on sharing and managing physical hardware efficiently.	ii) Focuses on on-demand access, scalability, and service delivery.
iii) Can operate within a private data center. Ex:- VMware ESXi.	iii) Can use virtualised or other infrastructure to provide services. Ex:- AWS EC2.

(4) A data center is a facility containing servers, storage systems, networking equipment, and supporting infrastructure used to process, store, and manage data.

(1) Computing : Hosts VMs, applications, and cloud workloads.

(2) Storage : Store databases, files, backups, and other digital info.

(3) Networking : Connects servers and enables communication between users and cloud services.

(4) Security :- Uses access controls, firewalls, and monitoring to protect systems and data.

(5) Availability :- Uses redundant power supplies, cooling systems and network connections to reduce downtime.

(6) Scalability :- Allows providers to add computing and storage capacity as demand increases.

(5)

Model	Full-form	Description
IaaS	Infrastructure as a service	Provides virtual machines, storage, and networking. Users manage the operating system and applications. Ex :- Amazon EC2.
PaaS	Platform as a service	Provides a platform for developing, testing, and deploying applications without managing the underlying infrastructure. Ex :- Google app engine.
SaaS	Software as a service	Provides ready-to-use software over the internet, usually through a browser or application. Ex :- Google workspace.
XaaS	Anything as a service	A broad term for delivering different IT resources and capabilities as services over a network. Ex :- Twilio